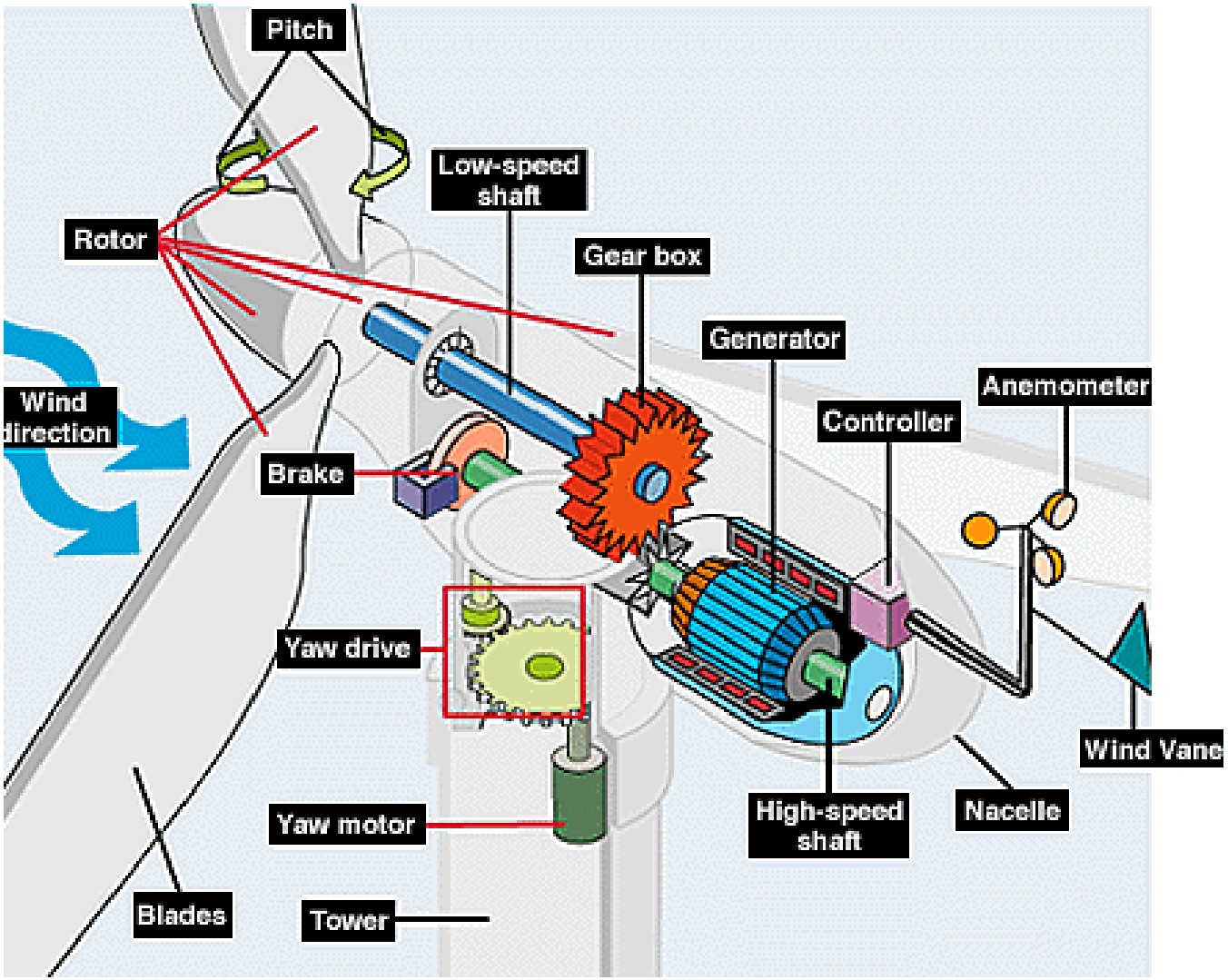
Obnovljivi izvori energije

Predavanje 12

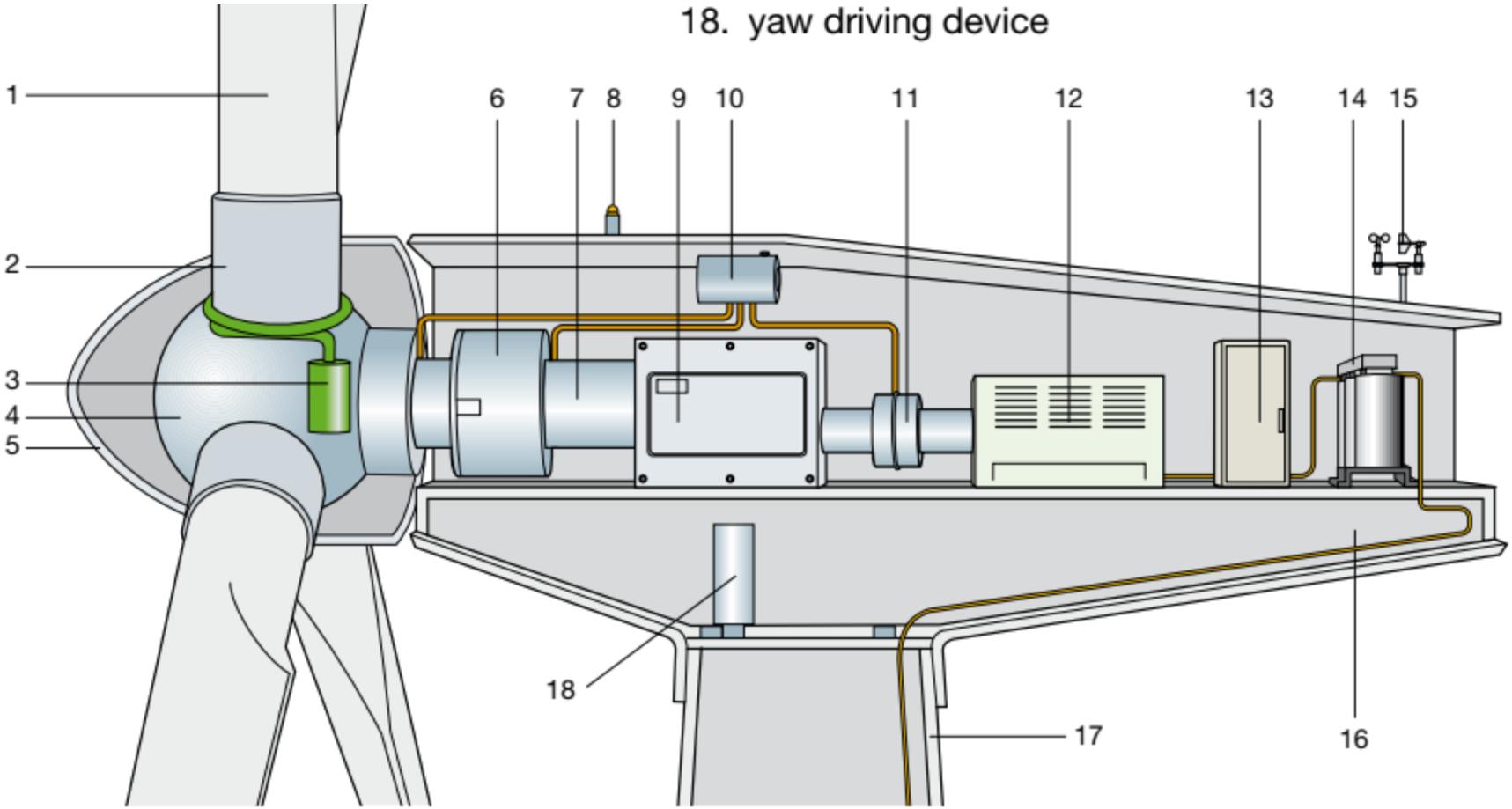
Princip konstrukcije vjetroagregata



Jedan realni primjer

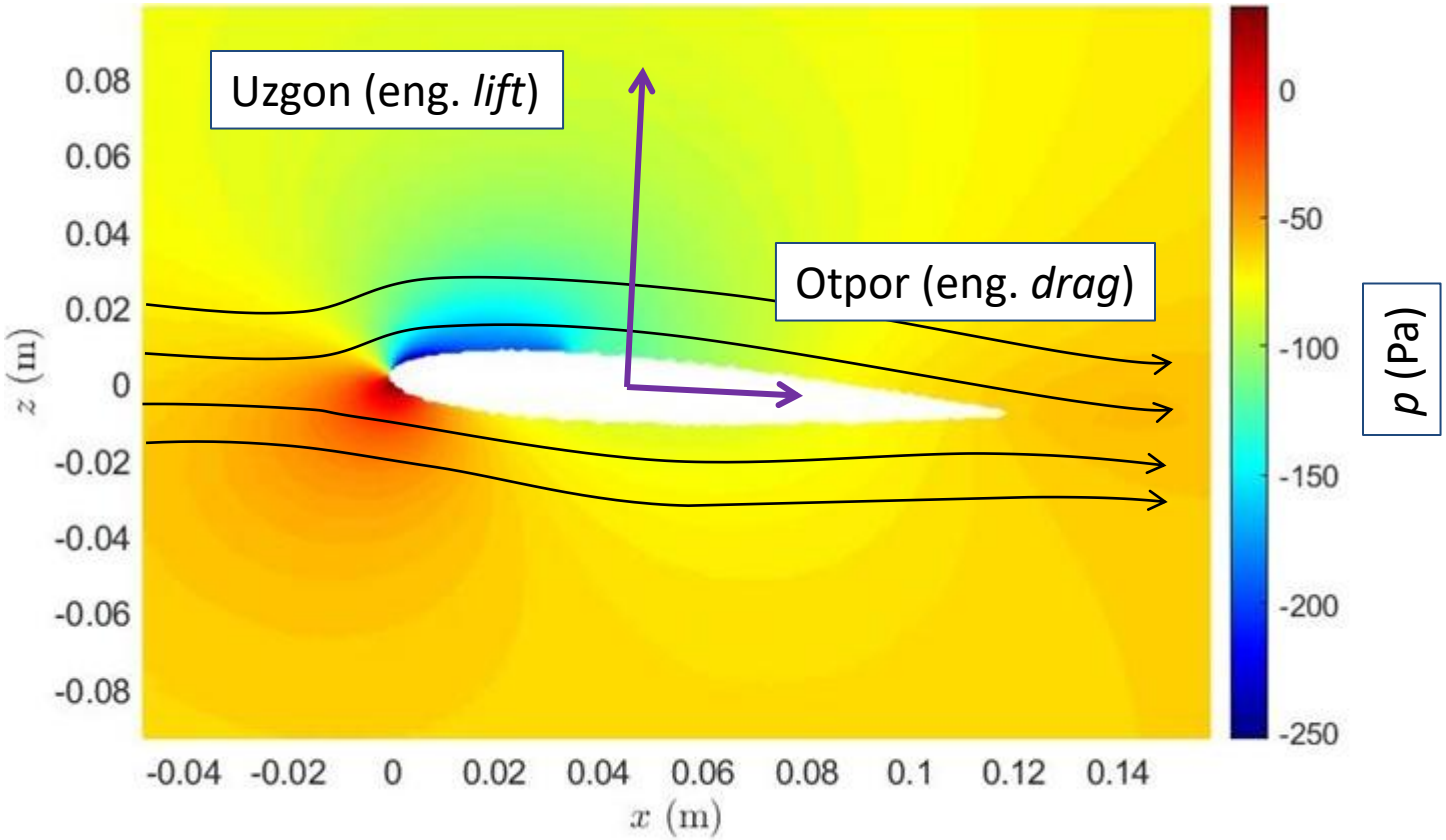
1 Hub/pitch panel
2 Pitch accumulators
3 Pitch/blade lock
4 Pitch actuator
5 Rotor lock system
6 Lubrication system
7 Through gearbox assembly
8 Hydraulic yaw system
9 Low speed brakes (gearless design)
10 Yaw brakes
11 Gearbox cooling system
12 Rotor and yaw brake system
13 Highspeed brakes
14 System for hatch opening
15 Hydraulic pitch power pack
16 Generator cooling system
17 Converter cooling system
18 Coolers with or without fan and rotor
19 Cooler

1. blade
2. blade support
3. Pitch angle actuator
4. hub
5. spinner
6. main support
7. main shaft
8. aircraft warning lights
9. gearbox
10. mechanical brakes
11. hydraulic cooling devices
12. generator
13. power converter and electrical control, protection and disconnection devices
14. anemometers
15. transformer
16. frame of the nacelle
17. supporting tower
18. yaw driving device



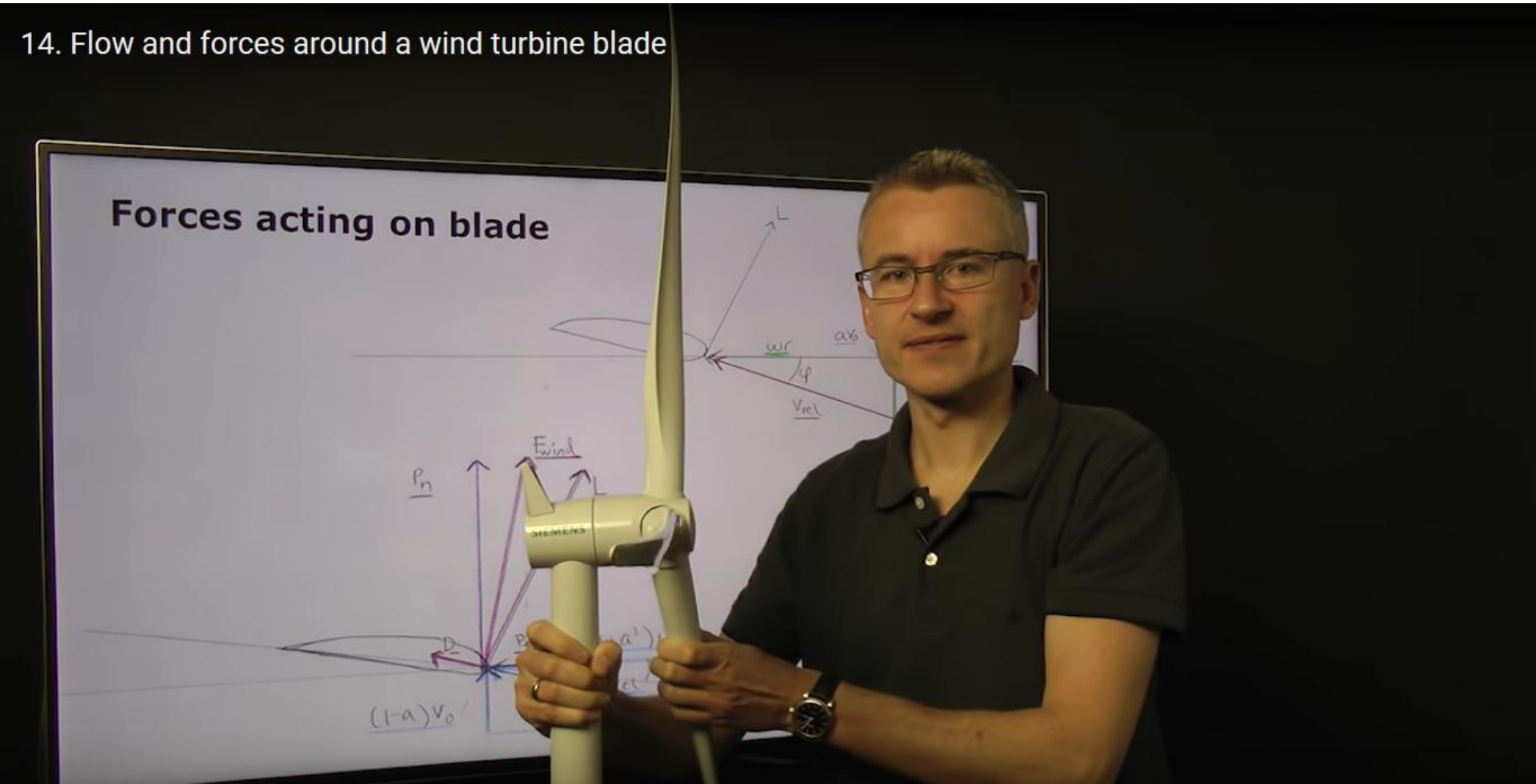
Turbina

- Osnovi aerodinamike turbinskih lopatica



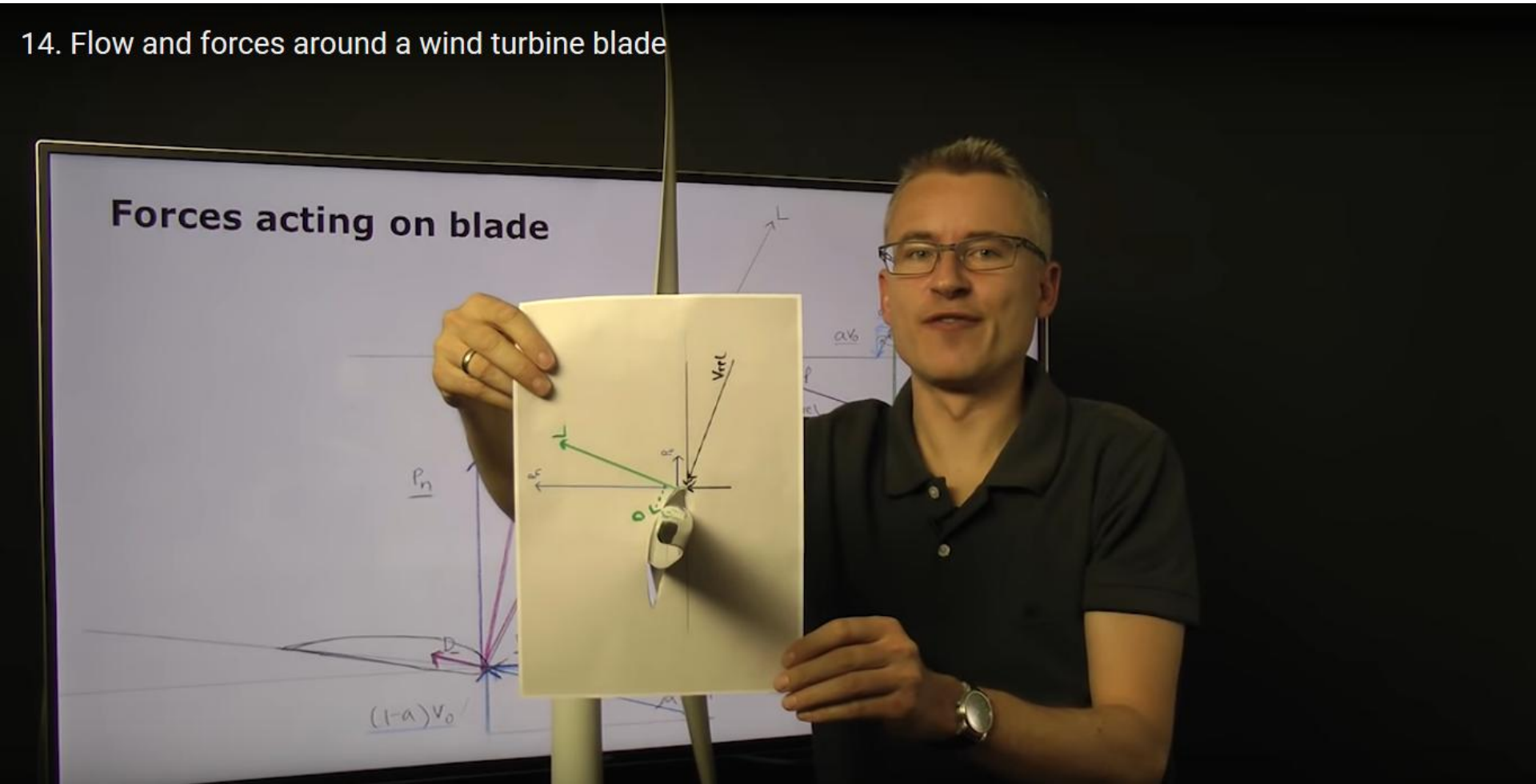
Turbina

- Osnovi aerodinamike turbinskih lopatica



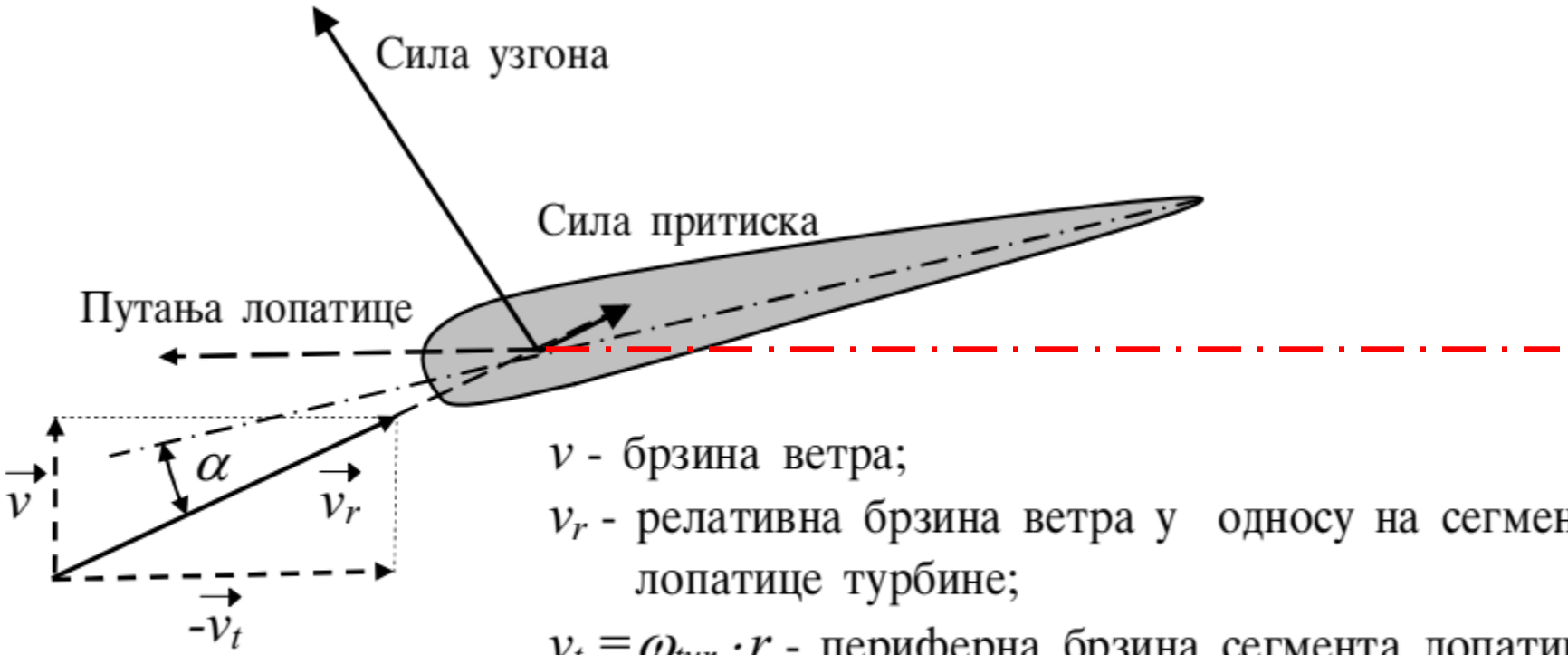
Turbina

- Osnovi aerodinamike turbinskih lopatica



Turbina

- Osnovi aerodinamike turbinskih lopatica



v - брзина ветра;

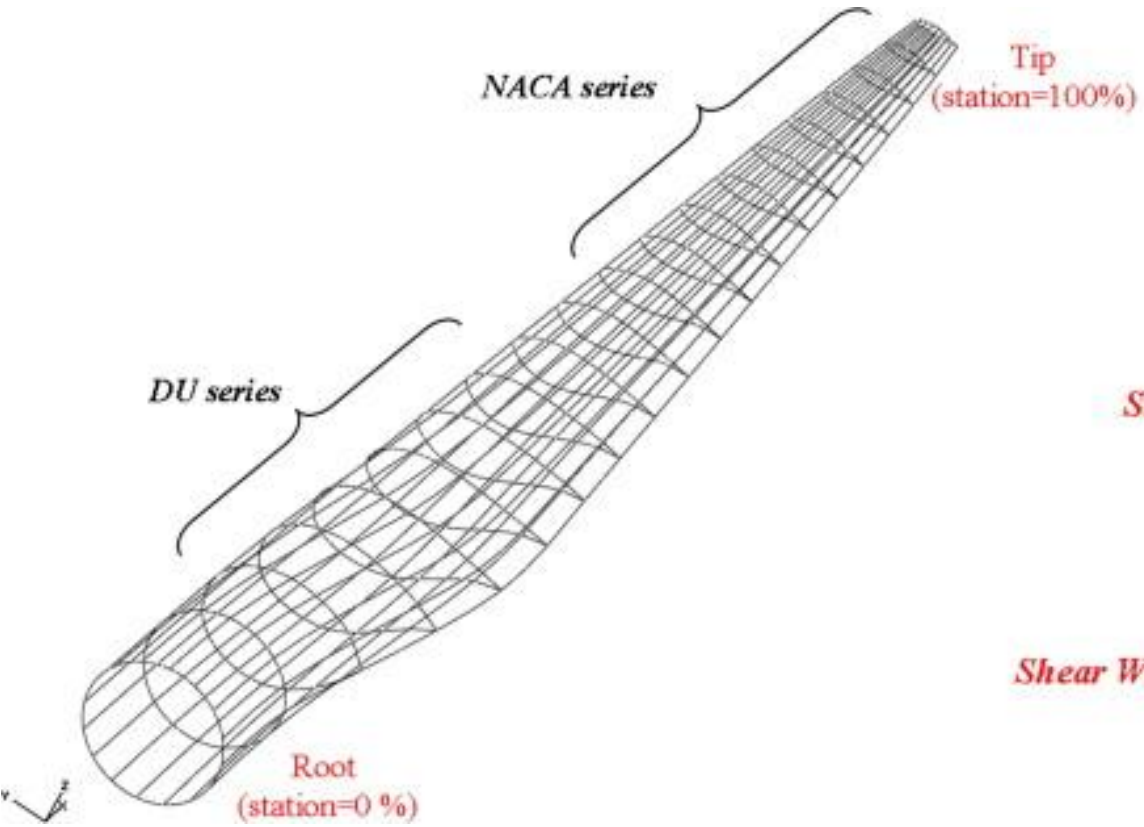
v_r - релативна брзина ветра у односу на сегмент лопатице турбине;

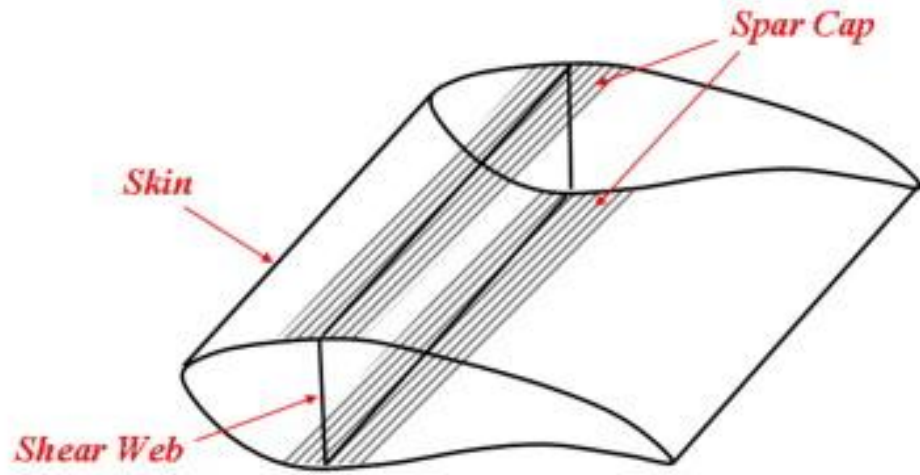
$v_t = \omega_{tur} \cdot r$ - периферна брзина сегмента лопатице турбине (r је растојање сегмента од аксијалне осе осовине турбине);

α - нападни угао.

Turbina

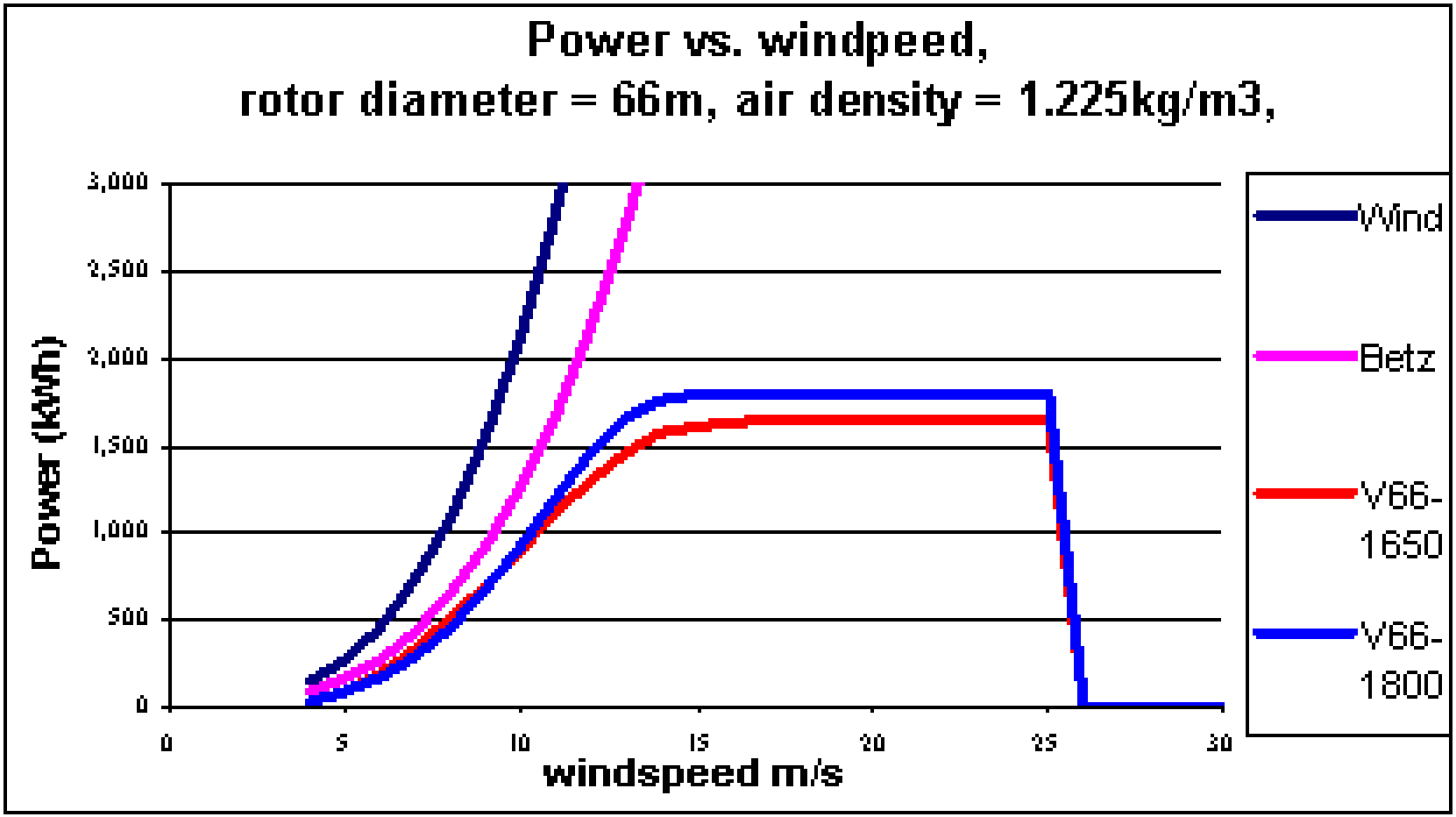
- Osnovi aerodinamike turbinskih lopatica





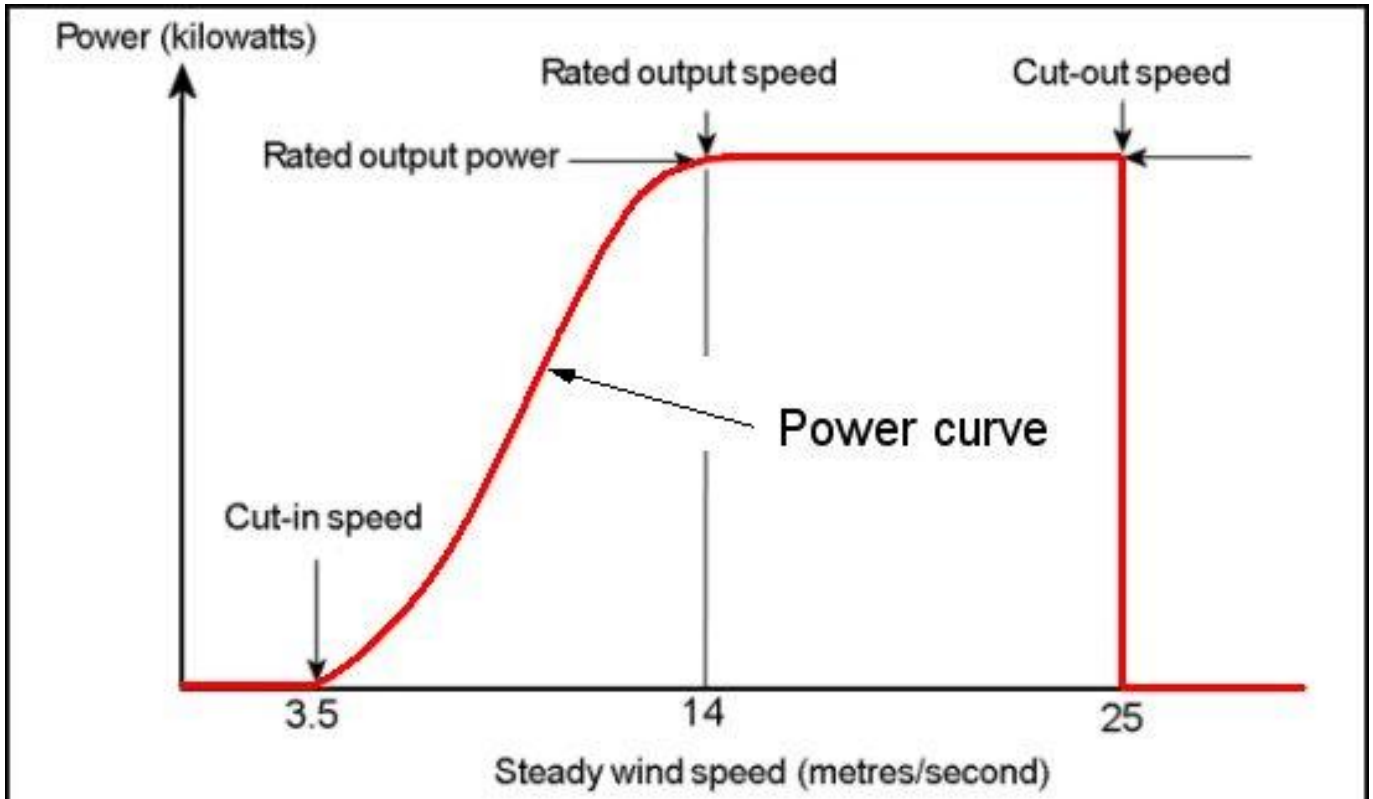
Karakteristika snage vjetroagregata

- Iskoristivi procenat od raspoložive snage vjetra



Tipična P = f(v) karakteristika

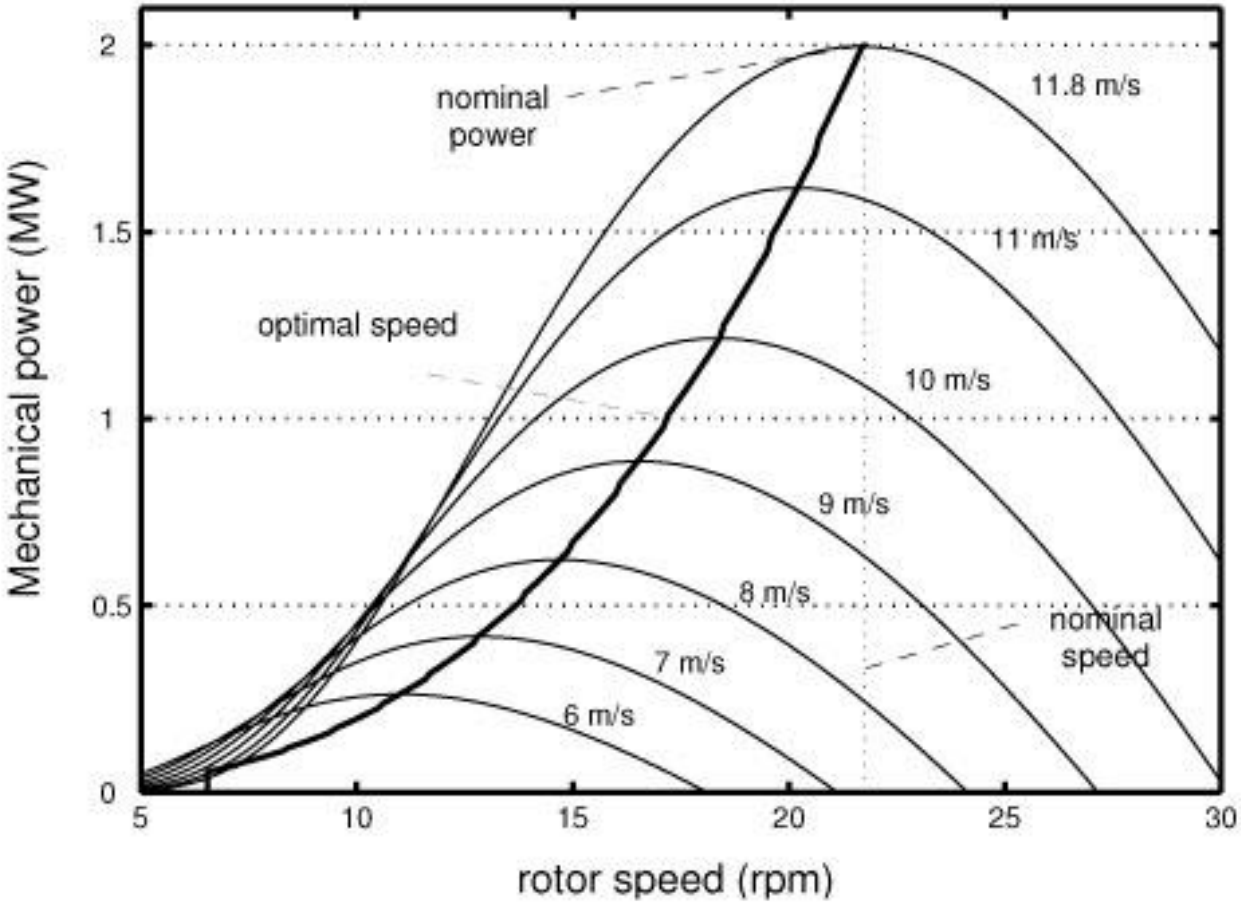
- Važne tačke P-v karakteristike vjetroagregata



Typical wind turbine power output with steady wind speed.

Regulacija snage vjetroagregata

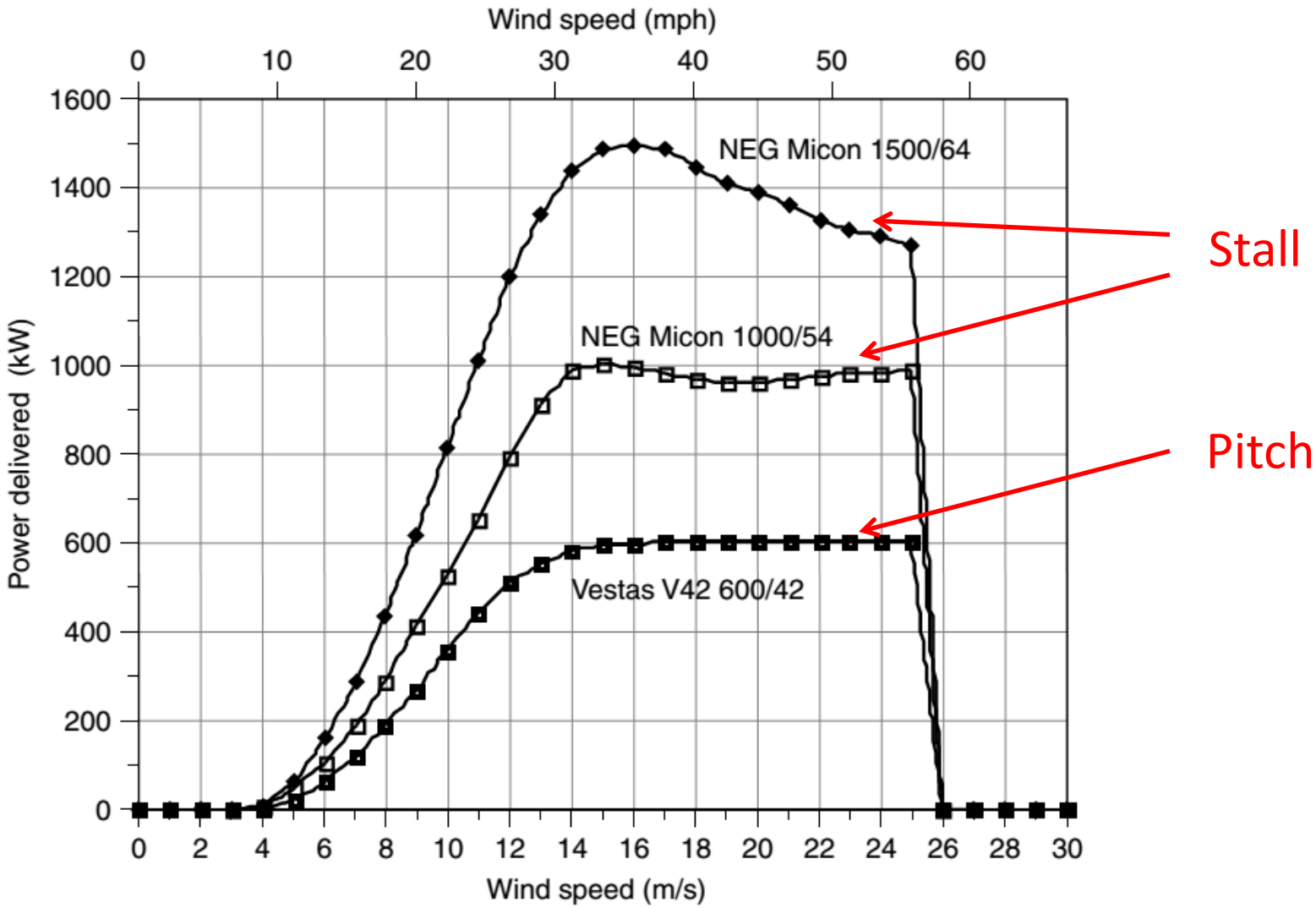
- Oblast $v < v_{nom}$



Regulacija snage vjetroagregata

- Oblast $v > v_{nom}$
  Snaga se održava na nominalnoj vrijednosti, s obzirom na ograničenje po instalisanoj snazi vjetrogeneratora.
  Metode regulacije:
  - Stall
  - Pitch
  - Combi stall

Regulacija snage vjetroagregata



Moguća proizvodnja energije

- Veza između godišnje proizvodnje vjetroagregata i prosječne brzine vjetra na nekoj lokaciji je nelinearna funkcija
- Godišnja proizvodnja se može dobro procijeniti pomoću tabelarnih proračuna (*Microsoft Excel* je odličan alat za to)

Windspeed (m/s)	Power (kW)	Probability $f(v)$	Hrs/yr at v	Energy (kWh/yr)
0	0	0.000	0	0
1	0	0.032	276	0
2	0	0.060	527	0
3	0	0.083	729	0
4	33	0.099	869	28,683
5	86	0.107	941	80,885
6	150	0.108	946	141,929
7	248	0.102	896	222,271
8	385	0.092	805	310,076
9	535	0.079	690	369,126
10	670	0.065	565	378,785
11	780	0.051	444	346,435
12	864	0.038	335	289,551
13	924	0.028	243	224,707
14	964	0.019	170	163,779
15	989	0.013	114	113,101
16	1000	0.008	74	74,218
17	998	0.005	46	46,371
18	987	0.003	28	27,709
19	968	0.002	16	15,853
20	944	0.001	9	8,709
21	917	0.001	5	4,604
22	889	0.000	3	2,347
23	863	0.000	1	1,158
24	840	0.000	1	554
25	822	0.000	0	257
26	0	0.000	0	0
			Total:	2,851,109

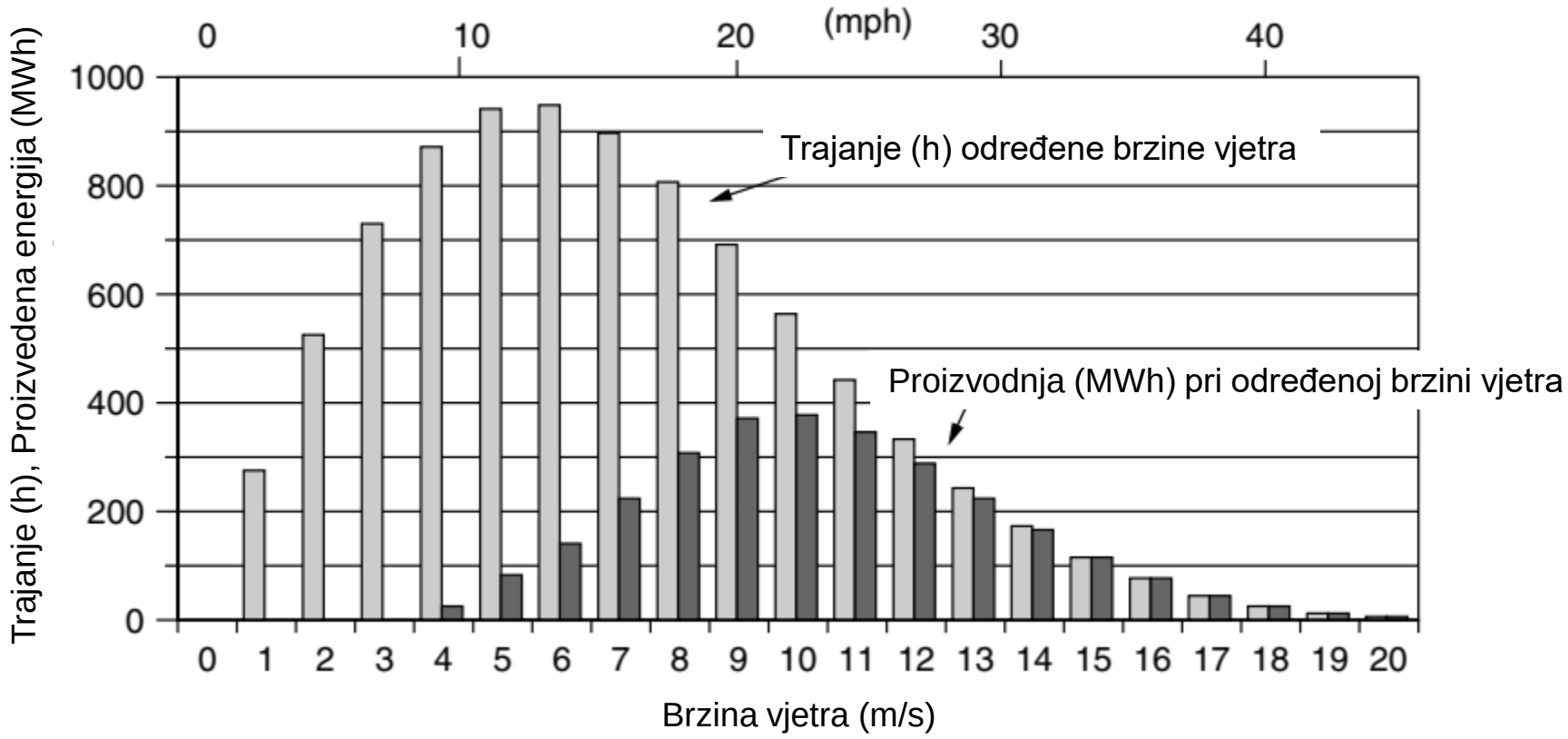
Podaci od proizvođača

Meteo podaci

Rezultat

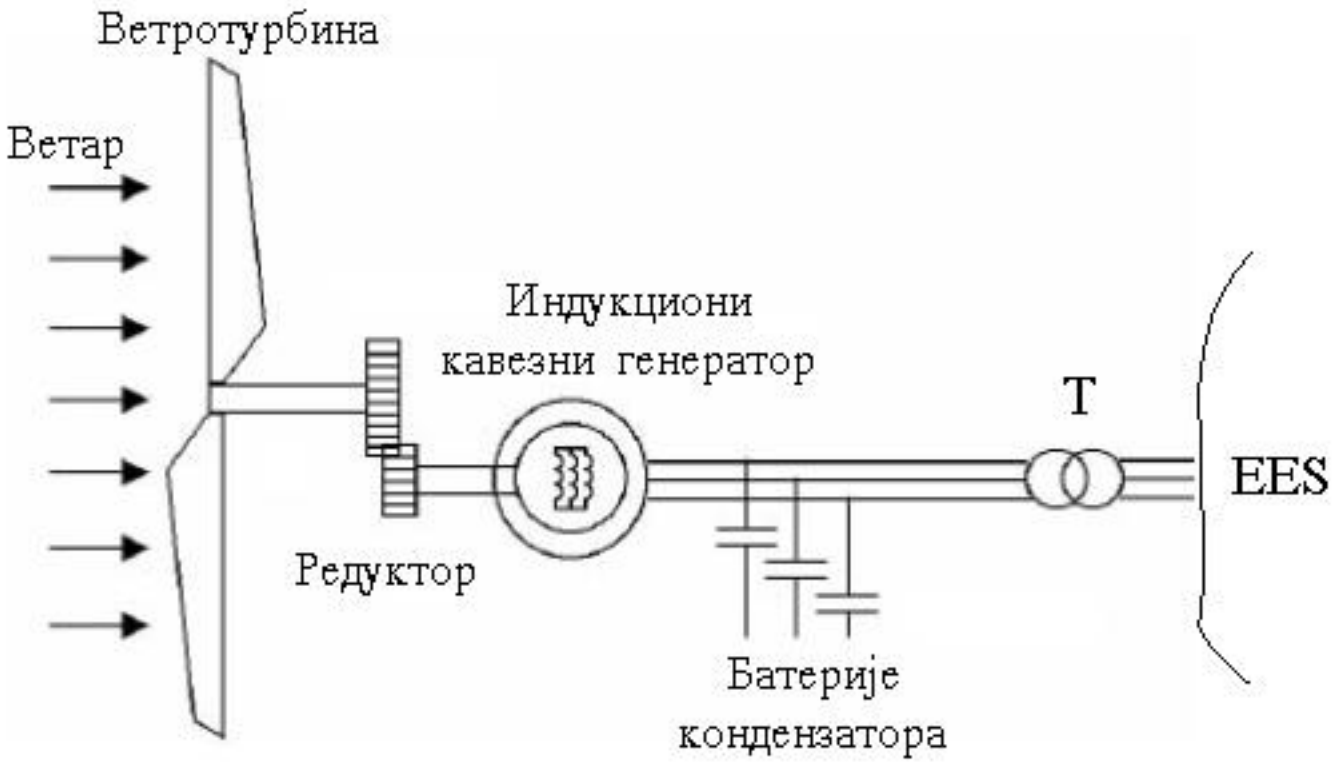
Moguća proizvodnja energije

- Dijagramski prikaz proizvedene energije u funkciji brzine vjetra



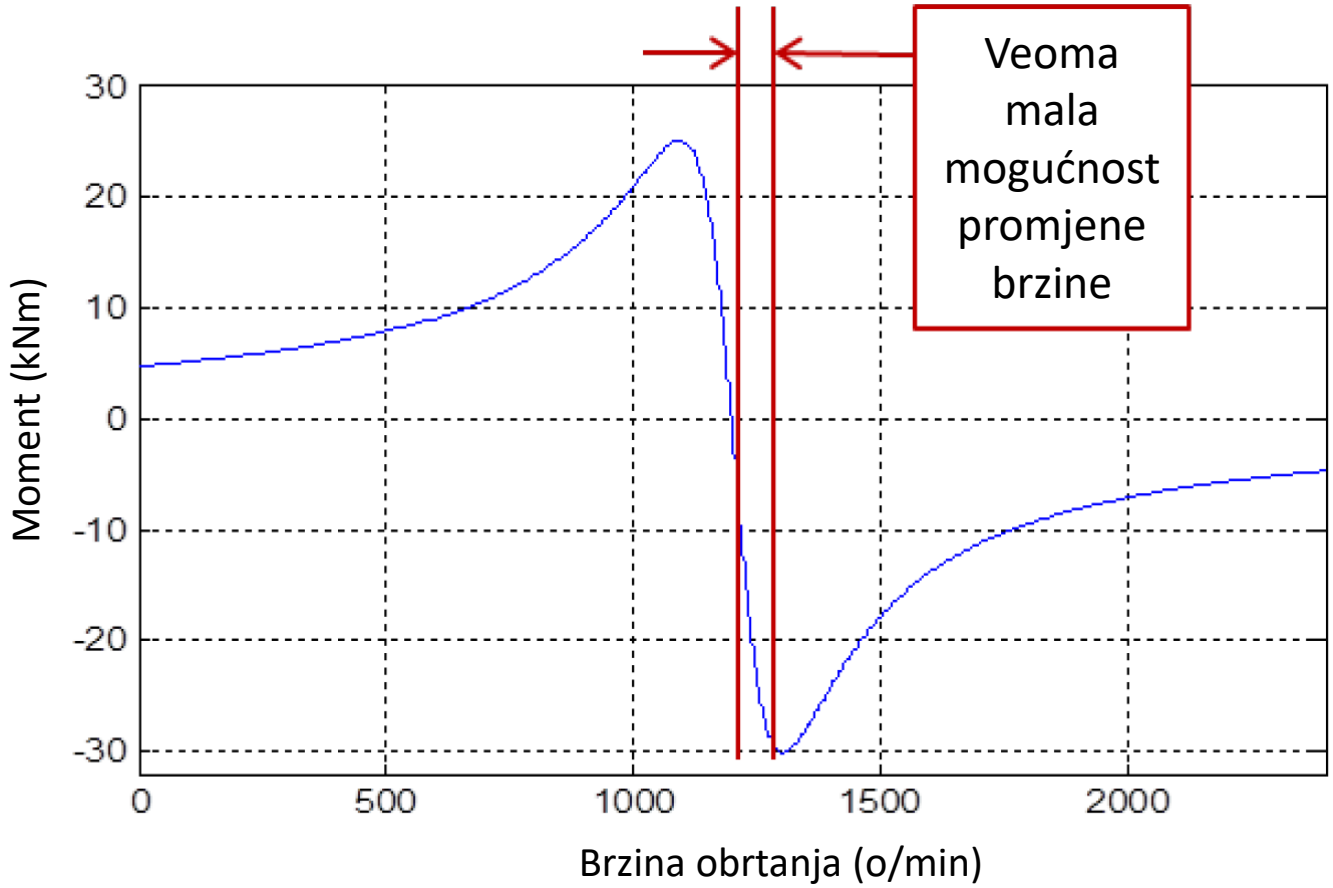
Koncepti konverzije energije

- Vjetroagregat sa asinhronim kaveznim vjetrogeneratorom



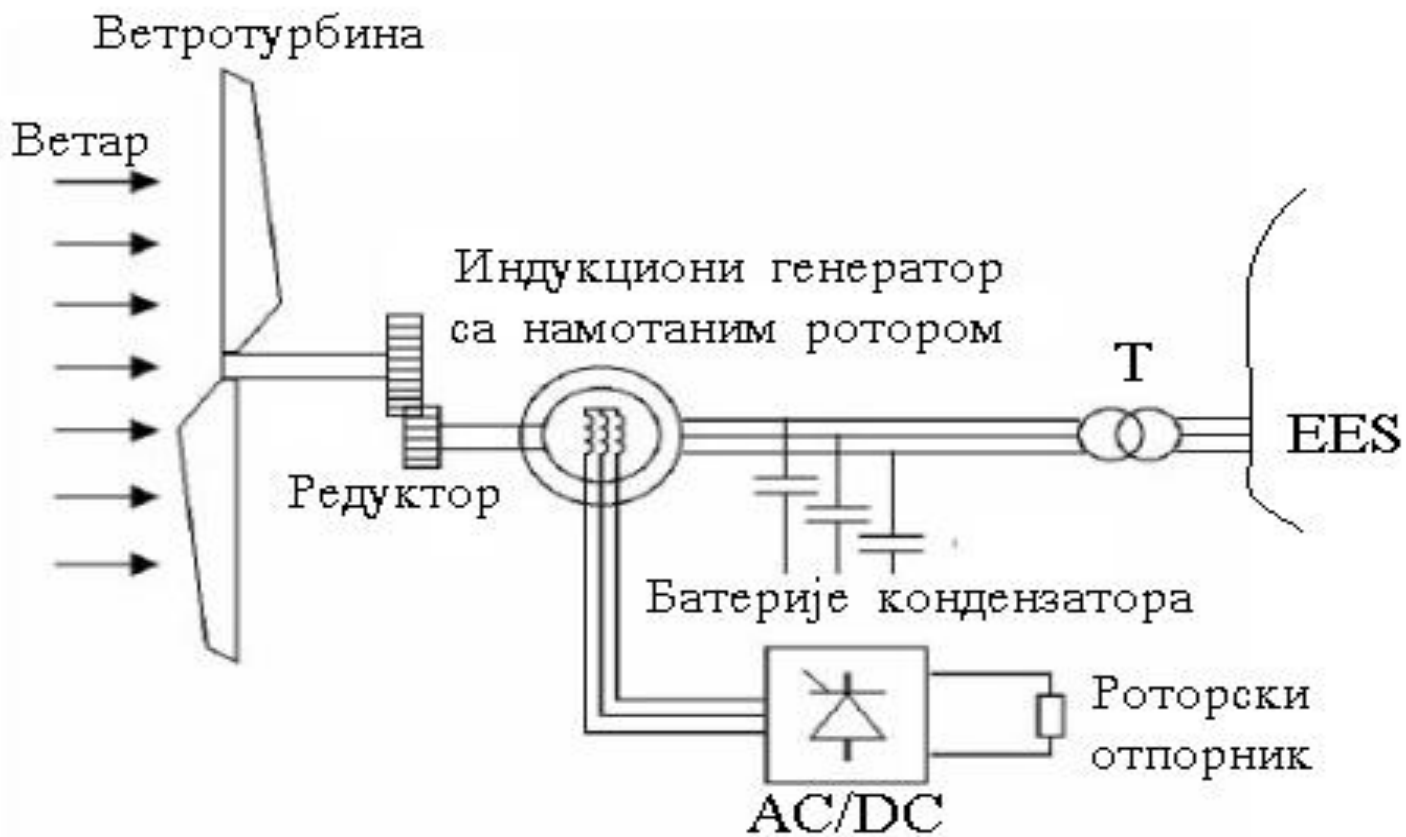
Koncepti konverzije energije

- Karakteristika kaveznog asinhronog generatora



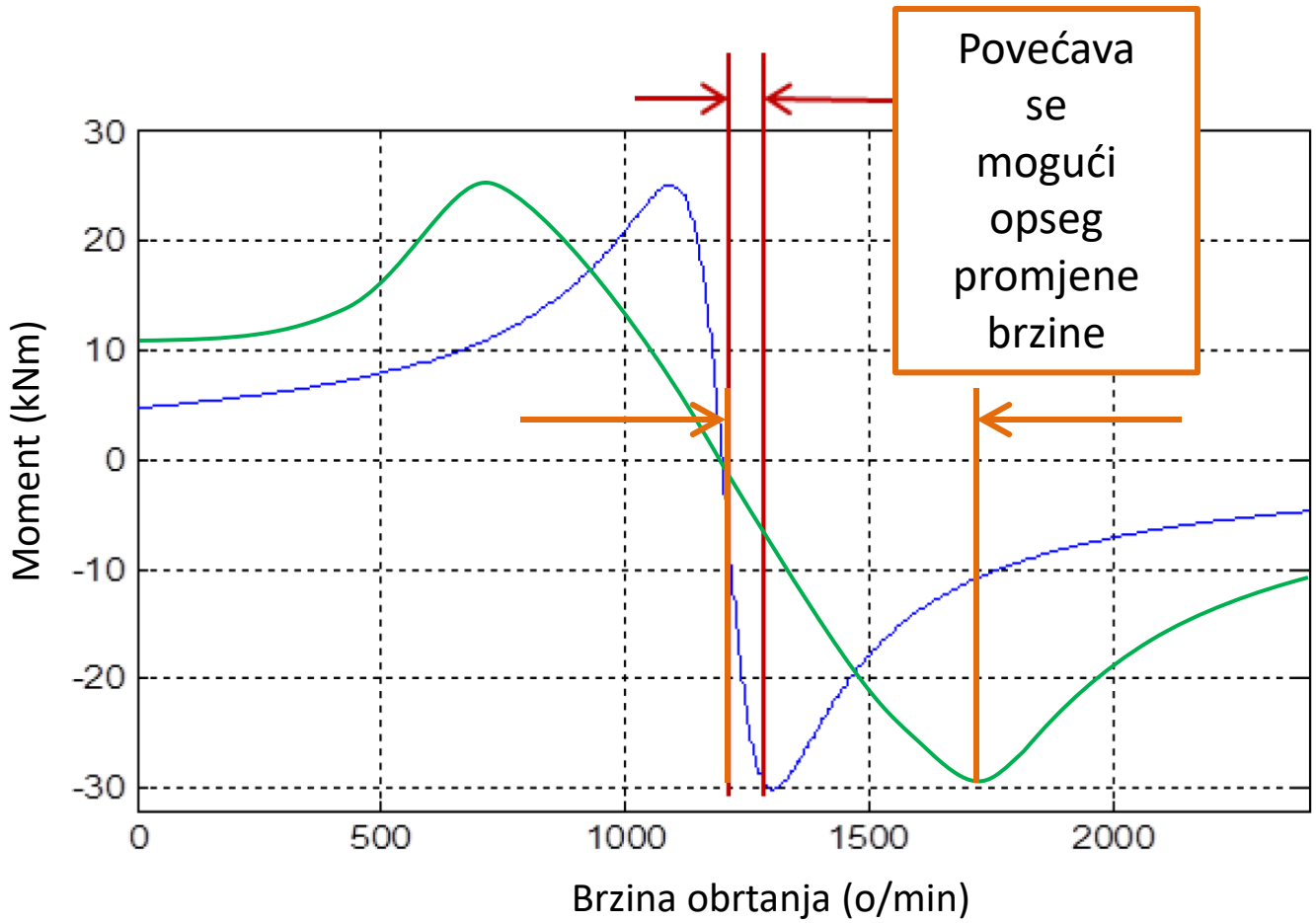
Koncepti konverzije energije

- Vjetroagregat sa asinhronom mašinom sa namotanim rotorom kao vjetrogeneratorom



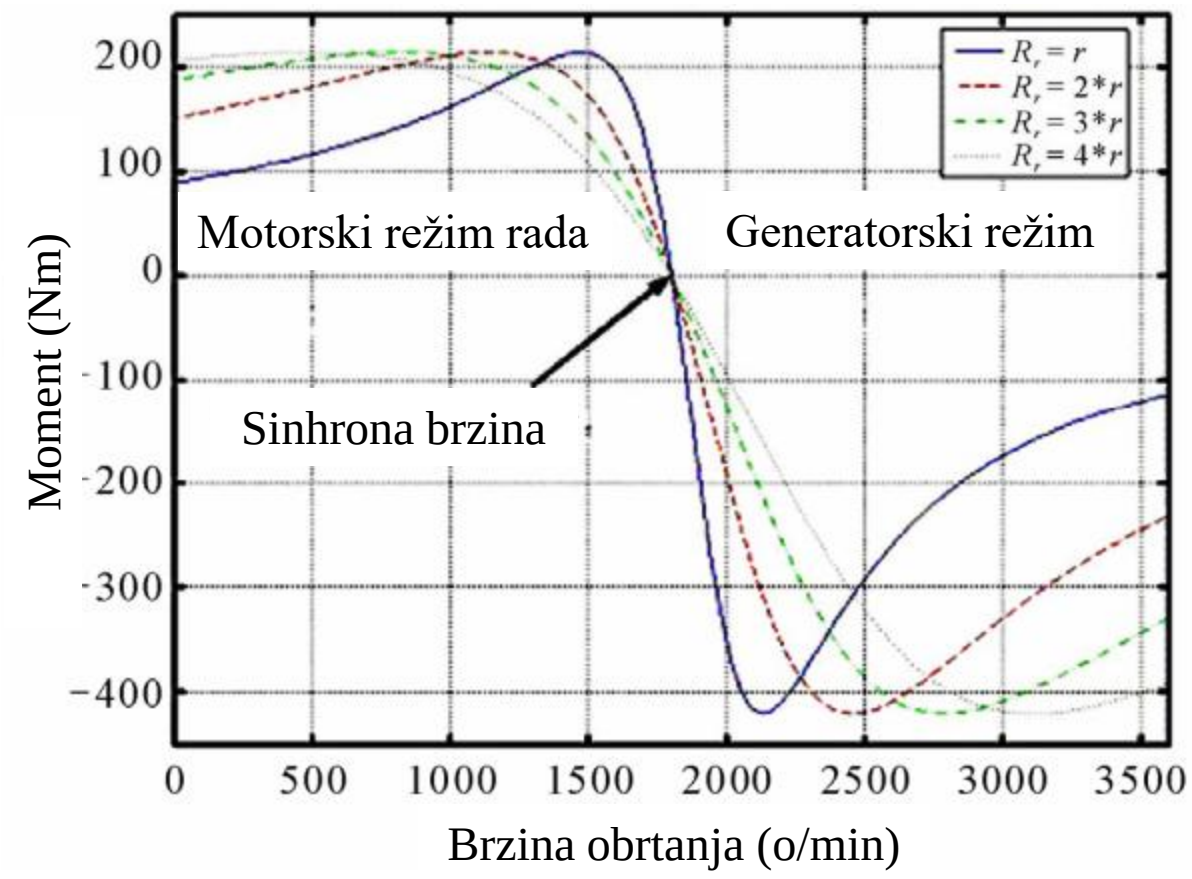
Koncepti konverzije energije

- Karakteristika kaveznog asinhronog generatora



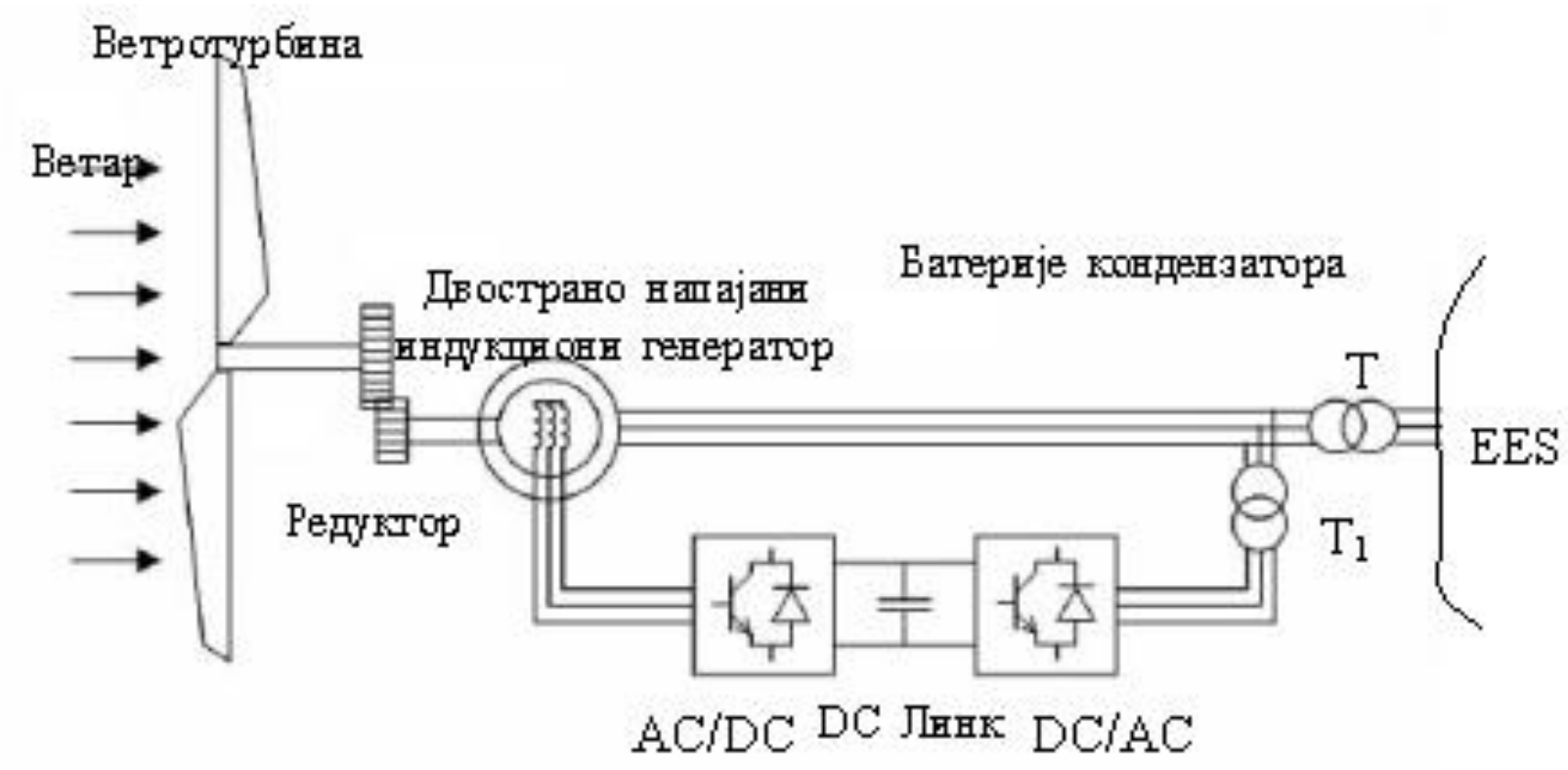
Koncepti konverzije energije

- Karakteristika kaveznog asinhronog generatora



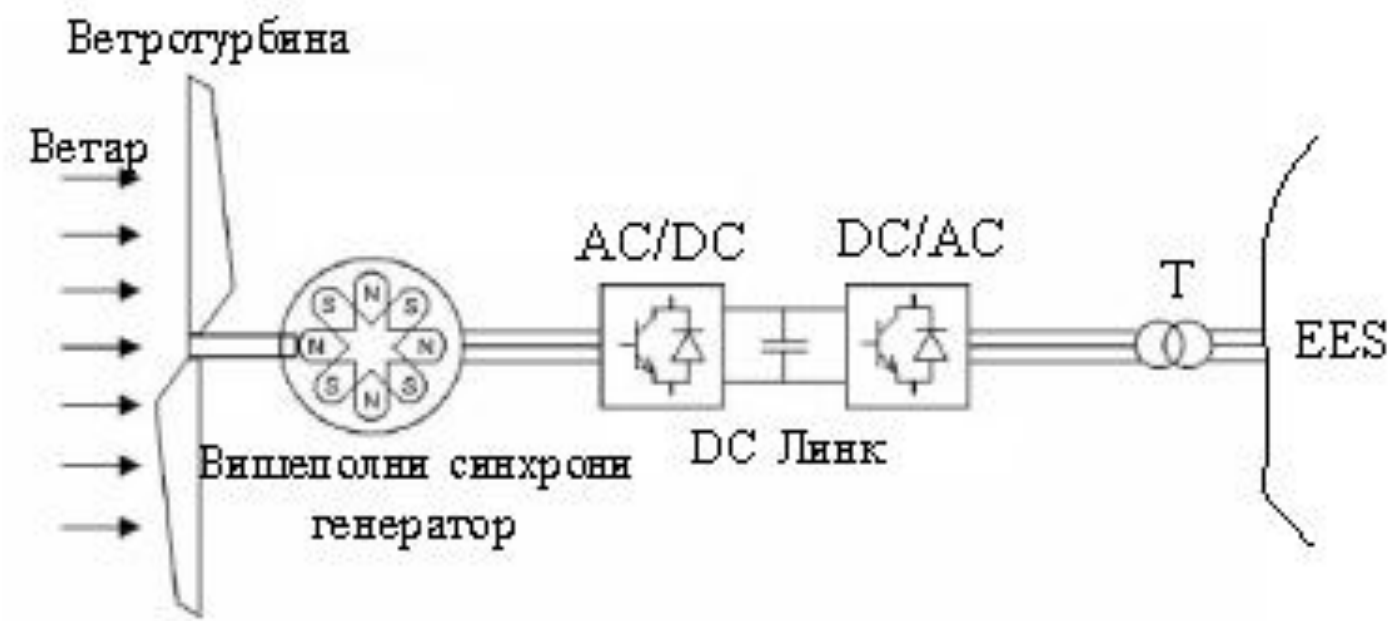
Koncepti konverzije energije

- Vjetroagregat sa vjetrogeneratorom - dvostrano napajanom asinhronom mašinom



Koncepti konverzije energije

- Vjetroagregat sa vjetrogeneratorom - mnogopolnom sinhronom mašinom sa permanentnim magnetima



Vjetroelektrana

- Više vjetroagregata se integriše u zajednički "park" – vjetroelektranu
- Neki od razloga za to su:
  - Smanjuju se troškovi izgradnje na tom prostoru
  - Pojednostavljuje se vezivanje na prenosnu mrežu
  - Centralizuje se sistem kontrole i nadzora

Vjetroelektrana

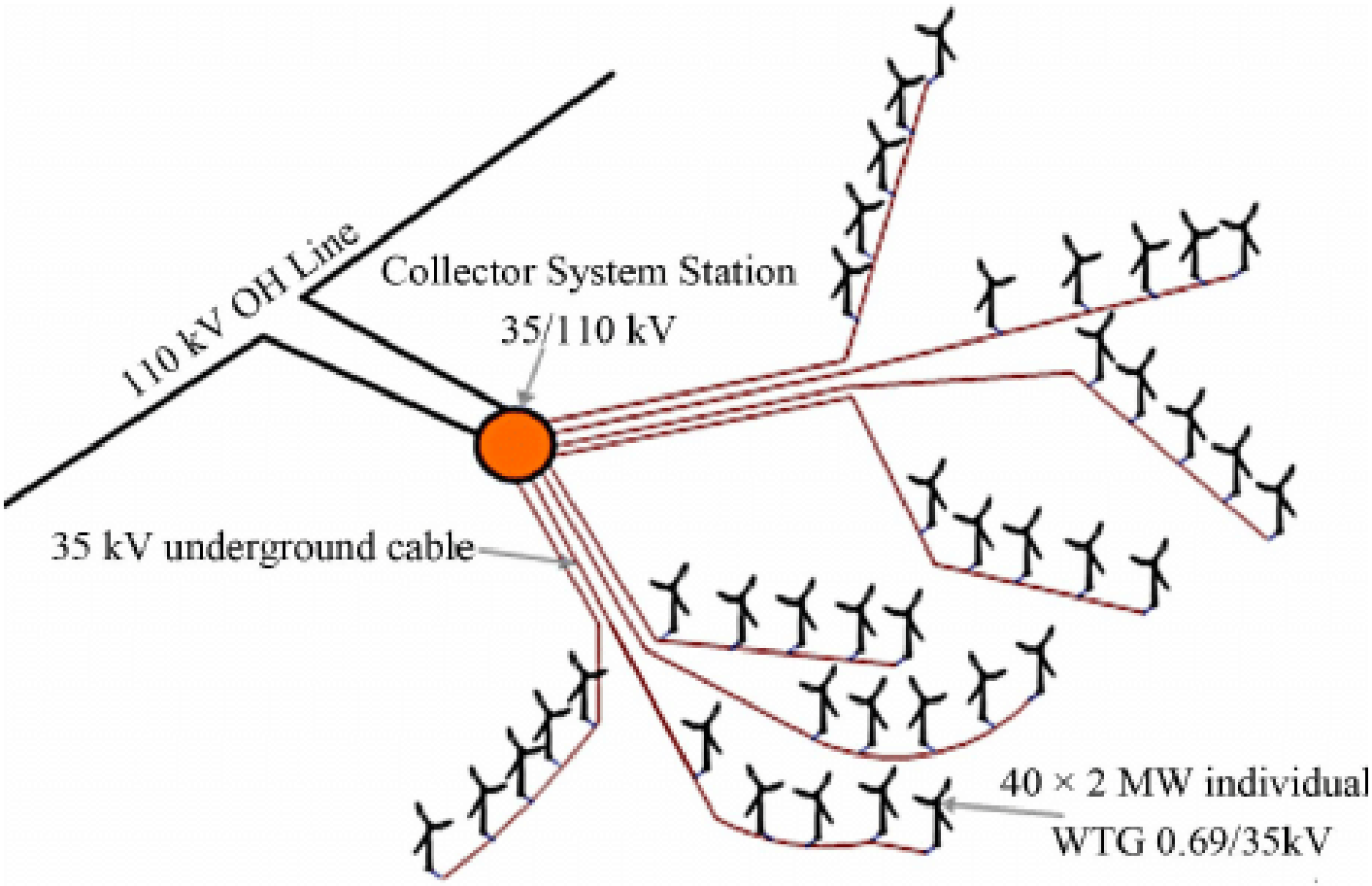
- Prema mjestu izgradnje
  - Elektrane na kopnu (onshore)
  - Elektrane na moru (offshore)
- Prema načinu isporuke energije
  - Elektrane povezane na mrežu
  - Elektrane za izolovanu proizvodnju energije

Vjetroelektrana na kopnu

Vjetroelektrana na moru

Vezivanje u postojeći EES

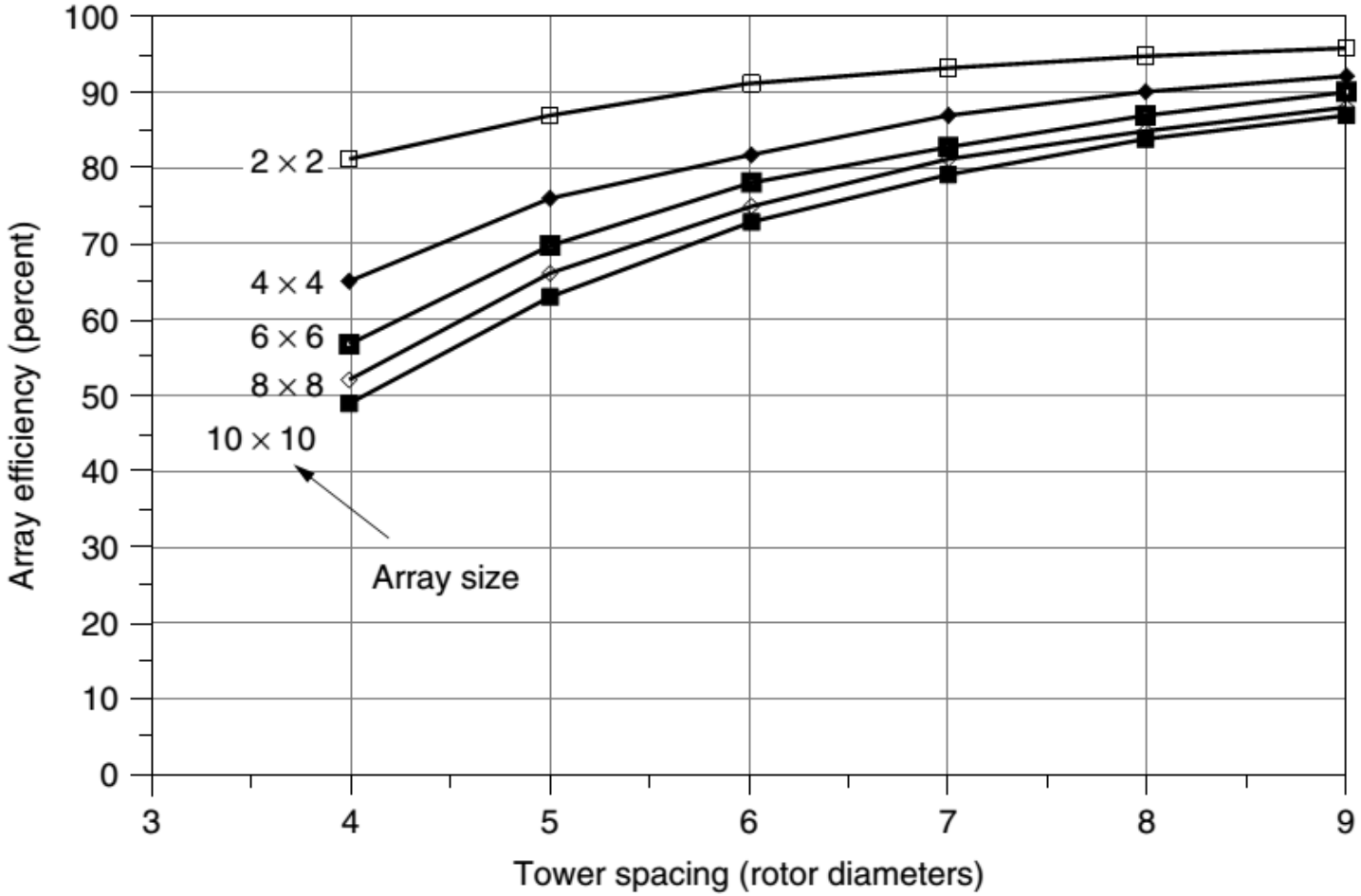
Šema vezivanja iz knjige



Efekat zavjetrine

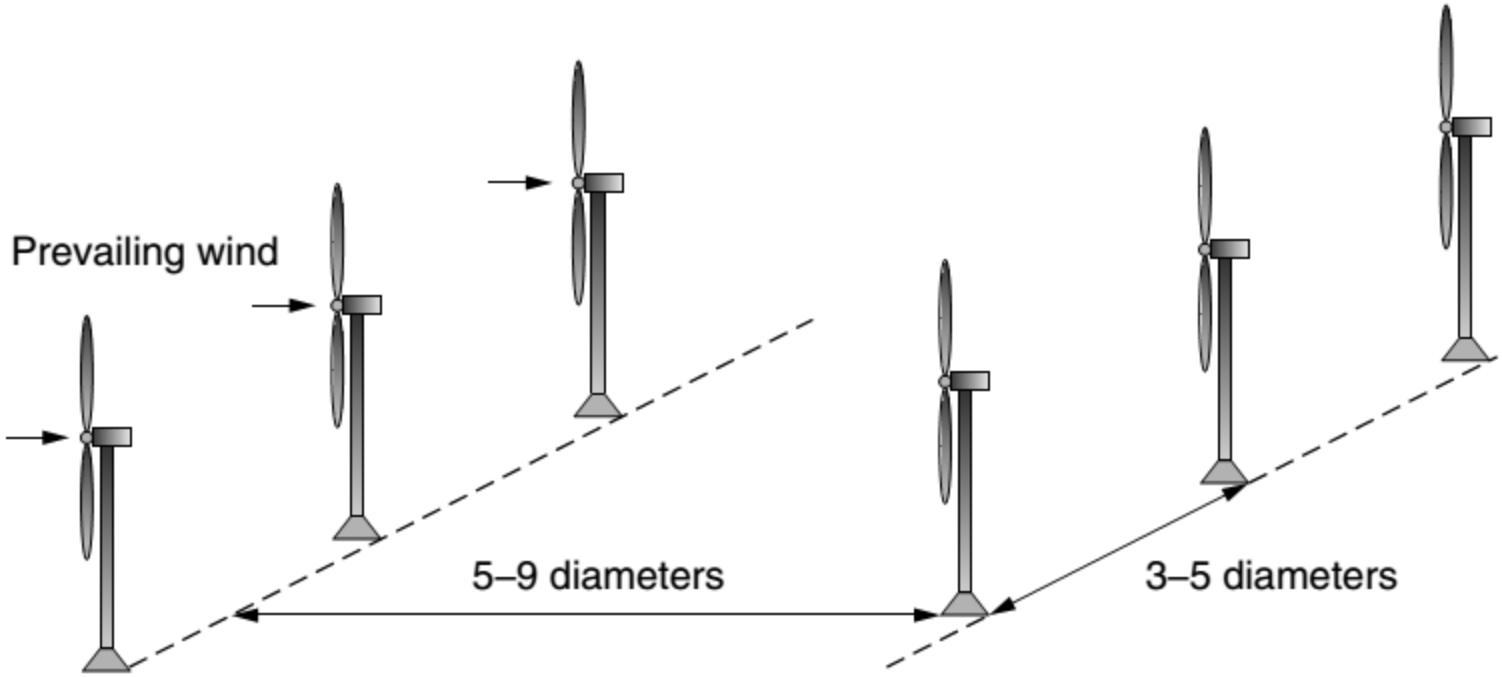
Efekat zavjetrine

- Javlja se problem međusobnog uticaja



Efekat zavjetrine

- Kompromisni razmaci između vjetroturbina



Ekvivalentna cijena energije

- Identičan metod korišćen za određivanje cijene kWh kod fotonaponskih elektrana i prezentovan u knjizi “Solarna energetika” Količnik ukupnih diskontovanih troškova i diskontovane proizvodnje energije
- *Feed-in* tarifa postojala u Republici Srpskoj (reda 0,14 KM/kWh), ali je posljednjim izmjenama pravilnika ukinuta

Uticaj na okolinu

- Pozitivan uticaj
  - Izbjegava se upotreba drugih “prljavijih” tehnologija za proizvodnju električne energije
- Negativan uticaj
  - Uticaj na faunu (ptice)
  - Buka,
  - Vizuelno zagađenje,